

SM-3: K3 Spectral Theorem and the Koide Formula

600-Cell Standard Model Emergence Series

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Abstract

We prove that the Koide relation $K \equiv \Sigma m_i / (\Sigma \sqrt{m_i})^2 = 2/3$ for the three charged lepton masses follows from the adjacency spectrum of the colour cage base graph K_3 (the equilateral triangle $\{V_1, V_2, V_3\}$ of the tetrahedral cage), with all three supporting propositions derived from CPP axioms and prior results. (P1) The ZBW Hamiltonian $\hat{H} = \hbar\omega_0 A_{K_3}$ follows from C3 cage symmetry and SSV hopping. (P2) $m_i \propto |\psi_i|^2$ follows from the CPP DI-bit visit rate. (P3) Equal eigenstate occupation follows from DP Sea thermalisation in the high-temperature limit ($kT_P/\hbar\omega_0 \sim 10^{20}$). Under P1 and P3, with P2 derived from the CPP DI-bit flow rate dynamics, the eigenvalue ratio $\lambda_{\max}/|\lambda_{\min}| = 2$ forces the Lorentz modulation depth $\rho = \sqrt{2}$, from which $K = (1 + \rho^2/2)/3 = 2/3$ follows exactly. The result is specific to K_3 : only the three-colour cage ($N = 3$) gives $K = 2/3$. The theorem applies to *leptons only*; quarks carry additional strong-sector mass contributions that systematically break Koide (deviations of 10–27% observed). The Koide phase θ is not determined by this theorem; its derivation requires the electroweak sector (OP-SM-7d).

1 Background

The Koide formula (Koide, 1982 [1]) is an empirical constraint satisfied by the three charged lepton masses to 11 ppm:

$$K \equiv \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}. \quad (1)$$

No Standard Model derivation exists. This paper proves $K = 2/3$ from the spectral structure of the K_3 cage base graph in CPP.

Note on independence: The K_3 graph is not chosen to reproduce the Koide relation. It is identified as the colour cage base of the 600-cell tetrahedral cage in SM-1 [4] from geometric considerations alone — specifically, the three colour vertices $\{V_1, V_2, V_3\}$ of the tetrahedral quark cage, whose C3 symmetry is forced by the 600-cell lattice geometry. The Koide relation emerges as a consequence of that independently-motivated structure, not as a motivation for it.

Prerequisites used without proof here:

- **Theorem 1** (Charge quantisation, SM-1 [4]): the cage base $\{V_1, V_2, V_3\}$ has C3 symmetry; $\delta = 1/3$ (topological proof from completeness + C3).

- **Algebraic identity:** with the parametrisation $\sqrt{m_i} = A(1 + \rho \cos \phi_i)$ and C3 phases $\phi_i = \theta + 2\pi i/3$, one has $K = (1 + \rho^2/2)/3$. Hence $K = 2/3 \Leftrightarrow \rho = \sqrt{2}$.
- **sea_strength** ≈ 0.1780 derived from the 600-cell Voronoi geometry (SS-1, Theorem 6 [3]).

2 The Colour Cage Base Graph

Definition 2.1 (Colour cage base graph K_3). *The three colour vertices $\{V_1, V_2, V_3\}$ of the tetrahedral cage, with edges connecting every pair, form the complete graph K_3 (equilateral triangle). Its adjacency matrix is:*

$$A_{K_3} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Lemma 2.1 (K_3 adjacency spectrum). *The eigenvalues of A_{K_3} are $\lambda_{\max} = 2$ (multiplicity 1, bonding) and $\lambda_{\min} = -1$ (multiplicity 2, antibonding). The bonding eigenvector is $(1, 1, 1)/\sqrt{3}$.*

Proof. $A_{K_3}(1, 1, 1)^T = 2(1, 1, 1)^T$. For $v \perp (1, 1, 1)$: $(A_{K_3}v)_i = \sum_{j \neq i} v_j = -v_i$, so $A_{K_3}v = -v$. $\square$

Remark 2.1 ($K = 2/3$ is unique to $N = 3$). *For the complete graph K_N , the eigenvalues are $N - 1$ (once) and -1 ($N - 1$ times). Applying the same equipartition argument gives $\rho^2 = N - 1$ and*

$$K = \frac{N + 1}{2N}.$$

Only $N = 3$ gives $K = 2/3$. The three-colour structure of the cage base is the precise reason the lepton Koide ratio is $2/3$.

3 The Three Postulates

Proposition 3.1 (ZBW Hamiltonian — derived from C3 symmetry and SSV hopping). *The ZBW orbital Hamiltonian restricted to the cage base is:*

$$\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3},$$

where $\omega_0 \equiv \text{sea_strength} \times c/r_{\text{conf}}$. This follows from C3 cage symmetry (Theorem 1), equal K_3 edge lengths $a = 1/\varphi$, and the SSV hopping amplitude (Proposition 3).

Remark 3.1 (Adjacency matrix vs. Laplacian). *The graph Laplacian $L_{K_3} = 2I - A_{K_3}$ has eigenvalues 0 (once) and 3 (twice). The Koide-relevant ratio $\lambda_{\max}/|\lambda_{\min}|$ is the same under either the adjacency matrix or the Laplacian; the adjacency matrix is the natural Hamiltonian for a nearest-neighbour hopping system, where off-diagonal elements represent hopping amplitudes between connected vertices. The result $K = 2/3$ is not an artefact of this choice.*

Proposition 3.2 (ZBW Born Rule — derived from P1 and P3). *The mass contribution of generation i is proportional to the squared ZBW wavefunction amplitude at colour vertex V_i : $m_i \propto |\psi_i|^2$. This is derived from P1 and P3 via the CPP DI-bit flow rate (Proposition 3).*

Proposition 3.3 (Thermal equipartition — derived from DP Sea thermalisation). *At thermal equilibrium with the DP Sea, all K_3 eigenstates are equally occupied, giving $|c_+|^2 = 1/3$ and $|c_-|^2 = 2/3$. Derived from DP Sea thermalisation (Proposition 3).*

Remark 3.2 (Physical motivation for P3). *In the high-temperature limit, each quantum state has equal occupation probability $1/g$ where g is the total number of states. For K_3 : $g = 3$ eigenstates, so each has probability $1/3$. The bonding sector contributes $1/3$; the two-dimensional antibonding sector contributes $2/3$. P3 is state-counting equipartition (equal weight per eigenstate), not energy equipartition.*

Remark 3.3 (Physical motivation for P1 and P2). *P1: The ZBW orbital hops between colour vertices via the electrostatic SSV interaction. For leptons (no colour charge), the amplitude is set by the DP Sea potential $V(r) = -\text{sea_strength} \times \hbar c/r$ at the confinement radius.*

P2: Each CP processes DI-bit flows at rate proportional to $|\psi_i|^2$ (the ZBW occupation probability at vertex i). Mass = stored ZBW energy $\propto$ DI-bit flow rate. P2 is the lepton-sector analogue of the Born rule (OP-QM-1).

Derivation of the ZBW Hamiltonian from C3 Symmetry and SSV Hopping

Proposition 3.4 (C3 symmetry forces $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$). *Given the C3 cage symmetry (Theorem 1) and equal K_3 edge lengths $a = 1/\varphi$, the ZBW Hamiltonian takes the form $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$ with $\hbar\omega_0 = \text{sea_strength} \times \hbar c/r_{\text{conf}}$.*

Proof. Part A — Uniformity (C3 symmetry). The rotation $\rho : V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_1$ is an exact isometry (Theorem 1). Any Hamiltonian commuting with ρ has equal off-diagonal elements $t_{12} = t_{23} = t_{31} \equiv t$ and equal on-site energies. Setting the energy origin to zero gives:

$$\hat{H}_{\text{ZBW}} = t A_{K_3}. \quad (2)$$

Part B — Amplitude $t = \hbar\omega_0$ (SSV hopping). For leptons, the hopping amplitude is set by the electrostatic SSV potential at the confinement radius:

$$t = \frac{\text{sea_strength} \times \hbar c}{r_{\text{conf}}} \equiv \hbar\omega_0 \approx 87.8 \text{ MeV}. \quad (3)$$

Conclusion. $\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$. □ □

Remark 3.4 (Consistency with SS-1 Theorem 6). *The hopping energy $\hbar\omega_0 = \text{sea_strength} \times \hbar c/r_{\text{conf}}$ is consistent with the DP binding energy $E_{\text{eDP}} = \hbar\omega_0/\varphi^2$ from SS-1 Theorem 6 [3]. The $1/\varphi^2$ factor arises from the Voronoi volume per vertex; $\hbar\omega_0$ is the total SSV energy at r_{conf} before the per-vertex projection.*

Derivation of the ZBW Born Rule from P1 and P3

Proposition 3.5 (DI-bit flow rate gives $m_i \propto |\psi_i|^2$). *Given P1 ($\hat{H}_{\text{ZBW}} = \hbar\omega_0 A_{K_3}$) and P3 (thermal equipartition), the lepton mass $m_i \propto |\psi_i|^2$.*

Proof. The ZBW orbital visits each colour vertex with frequency $\Phi_i = \omega_{\text{ZBW}} \times f_i$, where f_i is the fraction of Absolute Moments spent at V_i . The mass energy stored at vertex i per unit time is:

$$m_i c^2 = \hbar \Phi_i = \hbar \omega_{\text{ZBW}} f_i. \quad (4)$$

By P1, the lepton generation state for generation g is:

$$|\psi_g\rangle = c_+ |\phi_+\rangle + c_- e^{i \cdot 2\pi g/3} |\phi_-^{(g)}\rangle, \quad (5)$$

and the occupation at vertex i is $f_i = |\langle i | \psi_g \rangle|^2 = |\psi_i|^2$. Substituting: $m_i \propto |\psi_i|^2$. □ □

Remark 3.5 (P3 does double duty). *P3 simultaneously fixes $|c_-|^2/|c_+|^2 = 2$ (giving $\rho = \sqrt{2}$ and $K = 2/3$) and specifies the coherent generation state, making the ZBW trajectory ergodic and giving $f_i = |\psi_i|^2$. The theorem rests on **two** independent postulates (P1 and P3).*

Derivation of Thermal Equipartition from DP Sea Thermalisation

Proposition 3.6 (DP Sea thermalisation gives $|c_n|^2 = 1/3$). *The ZBW $K3$ resonator couples to the DP Sea thermal bath at the Planck temperature T_P . Since $kT_P/\hbar\omega_0 \approx 10^{20} \gg 1$, the Gibbs state reduces to the uniform mixture $|c_n|^2 = 1/3$ for each of the three $K3$ eigenstates, giving $|c_-|^2/|c_+|^2 = 2$.*

Proof. Step 1 (ZBW-DP Sea coupling). Every CP exchanges DI-bits with the DP Sea at each Absolute Moment. The ZBW orbital couples to the DP Sea via a Caldeira–Leggett system-bath coupling:

$$\hat{H}_{\text{coup}} = \sum_i \sum_k g_k |i\rangle\langle i| (b_k + b_k^\dagger). \quad (6)$$

Step 2 (Thermalisation). The relaxation time $\tau_{\text{relax}} \ll \tau_{\text{ZBW}}$ ensures rapid equilibration within each ZBW cycle. The equilibrium state is:

$$\rho_{\text{eq}} = \frac{e^{-\hat{H}_{\text{ZBW}}/kT_P}}{Z}. \quad (7)$$

Step 3 (High-temperature limit).

$$\frac{kT_P}{\hbar\omega_0} \approx 10^{20} \gg 1 \implies |c_n|^2 \rightarrow \frac{1}{3} \quad \text{for each eigenstate.}$$

Step 4 (Counting). K_3 has one bonding state and two antibonding states:

$$|c_+|^2 = \frac{1}{3}, \quad |c_-|^2 = \frac{2}{3}, \quad \frac{|c_-|^2}{|c_+|^2} = 2. \quad \square$$

□

Remark 3.6 (Physical meaning). *The ratio $|c_-|^2/|c_+|^2 = 2$ is the ratio of antibonding to bonding eigenstates of K_3 . At any temperature $kT \gg \hbar\omega_0$, this holds. Departures from $K = 2/3$ would only appear at $kT \lesssim \hbar\omega_0 \approx 88 \text{ MeV}$ — far above any energy scale accessible to stable leptons.*

4 The Theorem

Theorem 4.1 ($K3$ spectral origin of the Koide formula). *Under Postulates 3.1–3.3 and $C3$ cage symmetry (Theorem 1), the three charged lepton masses satisfy $K = 2/3$ exactly.*

Proof. Step 1 (Eigenstate structure, from P1). The ZBW eigenstates on K_3 are the bonding mode $|\phi_+\rangle$ at energy $2\hbar\omega_0$ and two antibonding modes at energy $-\hbar\omega_0$.

Step 2 (Occupation amplitudes, from P3). Thermal equipartition: $|c_+|^2 = 1/3$, $|c_-|^2 = 2/3$.

Step 3 (Modulation depth ρ).

$$\rho^2 = \frac{|c_-|^2}{|c_+|^2} = \frac{2/3}{1/3} = 2 \implies \rho = \sqrt{2}.$$

Step 4 (Koide K, algebraic identity). With C3 symmetry (phases $\phi_i = \theta + 2\pi i/3$) and $\rho = \sqrt{2}$:

$$K = \frac{1 + \rho^2/2}{3} = \frac{1 + 1}{3} = \frac{2}{3}. \quad \square$$

□

Corollary 4.2 (Common K3 origin of $\delta = 1/3$ and $K = 2/3$). *Both charge quantisation ($\delta = 1/3$, SM-1 Theorem 1) and the Koide formula ($K = 2/3$, Theorem 4.1) arise from the same triangle K_3 :*

Result	Value	K_3 structure used
Charge fraction δ	1/3	Combinatorial: 3 equal vertices, sum to 1
Koide ratio K	2/3	Spectral: eigenvalue ratio 2 : 1

5 Why Quarks Do Not Satisfy Koide

Remark 5.1 (Quarks: Koide broken by strong-sector mass). *The observed values are:*

$$K(d, s, b) = 0.731 \quad (9.7\% \text{ above } 2/3), \quad K(u, c, t) = 0.849 \quad (27\% \text{ above } 2/3).$$

Quarks carry qDP chain binding energy ($\sim 99\%$ of proton mass), inter-cage bonding, and cage-depth scaling. These strong-sector contributions break the K_3 spectral symmetry. Leptons carry no colour charge and no qDP chains; their masses are purely from K_3 ZBW modes, and Koide holds to 11 ppm. Theorem 4.1 is a lepton result, not a universal one.

6 Scope and Open Problems

Claim	Status	Notes
$K = 2/3$ (Koide formula)	Proved	Main theorem
$\rho = \sqrt{2}$ from K_3 spectrum	Proved	Step 3
C3 phase structure	Proved	From SM-1 Theorem 1
P1: $H = \hbar\omega_0 A_{K_3}$	Derived	C3 + SSV hopping
P2: $m_i \propto \psi_i ^2$	Derived	Via DI-bit ZBW visit rate
P3: Thermal equipartition	Derived	DP Sea thermalisation
Phase θ	Open	OP-SM-7d
Scale A	Calibrated to m_e	One free parameter
Individual masses m_μ, m_τ	Open	Need θ and A ; see SM-4

Open Problem 1 (OP-SM-7d: Derivation of the Koide phase θ). *Derive $\theta = 132.7323$ from CPP dynamics. Theorem 4.1 proves $K = 2/3$ but leaves θ undetermined. A structural theorem (proved in SM-4, Theorem 2 [5]) shows that no mechanism within the K_3 +SSV framework can select θ , because C3 symmetry leaves the antibonding subspace degenerate. The derivation of θ requires the electroweak sector (EW series, OP-EW-1).*

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